



IBM makes 2 nm chip

IBM has laid the claim to making the world's first 2 nm computer chip.
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Tech to save baby chickens

Sex identification for chicken is now a possibility, saving countless chicken lives.
<http://digit.in/may21-59>

MSI MEG Z590 ACE Motherboard

PCIe Gen 4.0 comes to Intel

MSI MEG Z590 ACE is the latest member of the ACE lineup of flagship motherboards from MSI and it's built for the latest 11th Gen Intel Core "Rocket Lake" desktop processor family. Let's take a closer look at the MSI MEG Z590 ACE.

BOARD LAYOUT

The MSI MEG Z590 ACE is packed to the hilt with features and has a very busy layout. Thankfully, the massive heatsinks spread across the length and breadth of the motherboard manage to hide it all and give it a cleaner look.

CPU Area and VRM

After taking the heatsinks off the board we get to take a gander at the VRM. The MSI MEG Z590 ACE uses the same Intersil ISL69269 controller as the Z490 ACE. For the power stages, it uses 16 Intersil ISL99390 90 Amp MOSFETs and for the SOC, it uses two Renesas RAA220075 75 Amp MOSFETs. MSI states that they've gone with a 16+2+1 power phase design on this motherboard. The Intersil ISL69269 controller can only handle up to 12 phases, so MSI is using doublers here to get the 16 phases for the Vcore and while the two phases for the SOC are not using any doublers. These doublers used are Renesas ISL617A and they can be seen on the rear of the board underneath the aluminium backplate. At the end of the day, the Vcore can use up to 1440 Amperes via this setup so you've got a pretty capable VRM set up here.

Expansion slots

The MSI MEG Z590 ACE has 3 x16 PCIe slots, 2 x1 PCIe slots and 4 NVMe slots. The first two of the x16 PCIe slots are from the CPU and the third one comes from the PCH. The



of the resizable BAR feature with AMD's RX 6000 cards and NVIDIA RTX 30 GPUs. More GPUs will receive access to the resizable BAR with future driver updates.

Price
₹47,990

Software

MSI includes their desktop MSI Centre application which is akin to their Dragon Centre software in laptops. The MSI Centre is a nice hub for monitoring all your system metrics and to play around with a few key features. The MSI Centre allows you to play around with MSI's Mystic Light RGB

system, Ethernet controller, monitor colour profiles, fan controller, set power plans or performance profiles and more.

VERDICT

The MSI MEG Z590 ACE can be seen as an incremental upgrade over the older MEG Z490 ACE

motherboard with new additions in the right places. The board is a bit more expensive than its predecessor and is built to cater to the needs of the PC enthusiast. We have fewer boards at launch with the Z590 platform considering that Alder Lake is just 6-8 months away. Nevertheless, this is a stop gap platform and possibly the last 14nm release before Intel moves into 10nm for the mainstream desktop platform.

—Mithun Mohandas

ones coming from the CPU support PCIe Gen 4.0 when a 11th Gen CPU is installed and the third x16 PCIe slot comes from the chipset so you only get PCIe Gen 3.0 on that.

With all the heatsinks removed, we see there are four 110mm M.2 slots on the MEG Z590 ACE. The first one is capable of PCIe Gen 4.0 when an 11th Gen CPU is installed while the remaining three only support PCIe Gen 3.0. The second and third M.2 slots from the top also support SATA drives and share bandwidth with the SATA ports on the right.

BIOS

MSI is still using their tried and tested Click BIOS 5 so there isn't much to talk about over here. There isn't much to add about MSI's Click BIOS 5 than what has been said before. The overall interface is quite straightforward for folks to understand. If you're looking for the Resizable BAR settings then head over to Settings>Advanced>PCI Subsystem Settings to find two options – Re-Size BAR and Above 4G memory. Turn these two on and you can make use



DESIGN.....	85
PERFORMANCE.....	90
VALUE FOR MONEY.....	75
FEATURE.....	82

SPECIFICATIONS

CHIPSET: Intel Z590 | MEMORY SUPPORT: 4x5600(OC) MHz upto 128 GB | LAN: Intel I225-V 2.5G Gigabit Ethernet | AUDIO: ALC 4082 + ESS SABRE9018Q2C | SATA: 6x SATA III | EXPANSION: 3x PCIe x16(x16+x0+x4) | DIMENSIONS (LXWXD): 305 mm x 244 mm | WARRANTY: 3 years.

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